Netflix Recommendation System

Two-Stage Collaborative Filtering:

100M ratings | 287,916 users | 16,560 movies | Best MAP@10 = 0.087 | Best RMSE=0.98

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Problem Overview

The Netflix Prize dataset is a sparse user-movie rating matrix (1–5 stars). We have two problems : Rating accuracy + Ranking quality

Rating Prediction (RMSE) (1-5 clip)

- Predict the exact movie rating for each (user, movie) pair
- Challenge: precision and cold-start for unseen users

Movie Recommendation (MAP@10)

- Retrieve & rank unwatched movies a user would rate ≥ 3.5
- Challenge: Items recommended should match the **exact movies** and in the **order,** the users prefer them **out of 17k movies available.**

40%

of test users were cold (never seen in training)

Cold-start dominates the evaluation set

An 80/20 temporal split was applied, with filters requiring ≥50 ratings per user and ≥100 per movie because other entries act as cold-start problems and add noise. This leaves 287,916 users , 16,560 movies, and 77M ratings. Even so **40% of users never appear in training**, forming the cold-start problem.

98.38%

Sparse data : user-movie pairs don't exist in training

Sparsity and Scale

With 98% empty pairs, giving ratings to cold users become tough. Also, there are 17k movies in the dataset so a user has 17k choices and predicting correct 10 recommendations for each becomes difficult with this scale.

Key insight: A model that predicts ratings well doesn't automatically rank movies well. We need a dedicated ranking objective.

Approach: Two-Stage Retrieve-and-Rerank Pipeline

STAGE 1

WALS — Retrieval & Rating Prediction

- Weighted Alternating Least Squares on 77M ratings
- Learns 128-dim user/item latent factors + bias terms + global mean (3.579)
- Warm score = μ + b_u + b_i + p_u · q_i
- Cold-start fallback: score = μ + b_i (item bias only)
- Output: top-2000 candidate movies per user for stage 2

```
==========================================
 Best Train RMSE (Full)     : 0.7189
 Best Test RMSE (Full)      : 1.0003
 MAP@10 (Full)          : 0.0160
------------------------------------------
 Warm Test RMSE (Personal) : 0.9390
 Warm MAP@10 (Personal)   : 0.0184
------------------------------------------
 Cold Test RMSE (Fallback) : 1.0842
 Cold MAP@10 (Fallback)   : 0.0025
==========================================
```

- **Warm users achieved lower RMSE than cold users**, as the model generates stronger embeddings with sufficient rating history. Cold users were not handled, leaving room for improvement via cold-start solutions outlined in the report.

▶

top-2000 candidates per user

STAGE 2

BPR — Reranking for Recommendations

- Bayesian Personalised Ranking on the 2000-item candidate pool
- 64-dim residual embeddings, warm-started from WALS item biases
- Score = softplus(scale) × (e_u·f_i + bias_i) + wals_w × norm(WALS)
- Curriculum hard-negative sampling: ratio = min(epoch/30, 0.36)
- Adam (lr=1e-3), early stopping patience 5 → top-10 final list

- **Cold users achieved higher MAP:** they were recommended popular movies which aligned well with their unknown preferences. Warm users, having personal tastes, saw lower MAP as recommending their expected top 10 out of 17k movies is tough.

```
Cohort [Warm] Results:
  Users evaluated: 212,166
  MAP@10:          0.0521
  NDCG@10:         0.1027
Eval Cold: 100%|██████████| 39284/

Cohort [Cold] Results:
  Users evaluated: 39,284
  MAP@10:          0.1319
  NDCG@10:         0.2188
Eval All: 100%|██████████| 251450/

Cohort [All] Results:
  Users evaluated: 251,450
  MAP@10:          0.0646
  NDCG@10:         0.1208
```

WALS optimises RMSE and gives a cheap, cold-start-friendly candidate set; BPR then optimises ranking quality directly over a much smaller pool (2,000 vs 16,560 items).

Approach: Two-Stage Retrieve-and-Rerank Pipeline

STAGE 1

Two Tower Model — Retrieval & Rating Prediction

- Dual PyTorch MLPs learning 32-dim User & Item representations
- Objective: Trained using Mean Squared Error (MSE) to predict 1-5 star ratings
- Incorporates user_bias, item_bias, and global_bias (gracefully handles Cold Users)
- Rapidly scores ~18k movies via dot-product of user & item representations
- Output: Filters catalog down to a highly relevant shortlist of 2,000 candidates

▶

top-2000 candidates per user

STAGE 2

LIGHTGBM — Reranking for Recommendations

- Gradient Boosting Decision Trees optimized for pairwise ranking (LambdaMART)
- Engineered 74 rich dense features per user-candidate pair
- Features: Two-Tower scores, raw 32-dim embeddings, and statistical baselines
- Recency Score: Cosine similarity to the user's most recently watched movie
- Rating Gap: Candidate's avg rating minus the user's historical avg rating
- Output: Learns complex non-linear interactions for top-10 final list

Roadmap to MAP@10 = 0.087

1. Two Tower - BPR loss

MAP@10 ~ 0.045

Linear ranker struggled with non-linear deep embeddings.

2. Two Tower- LightGBM (1k Candidates)

MAP@10 = 0.082

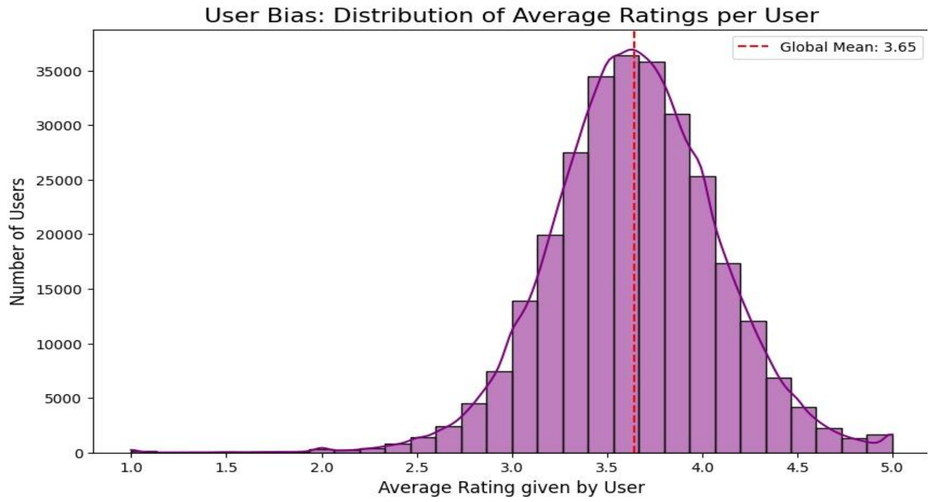
Massive surge. Recency score dynamically shifted recommendations.

3. Expansion (2k Candidates)

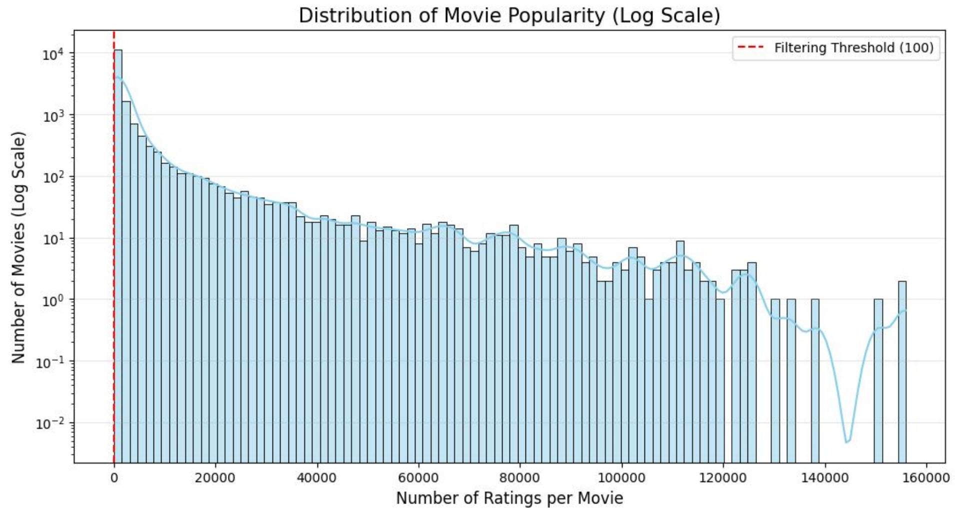
MAP@10 = 0.087 (Final)

Resolved 30M row OOM crashes by downcasting matrices to float16.

User Behaviour & Rating Distribution



Observation: The distribution shows a peak at 3 and 4 stars, with significantly fewer 1 and 2-star ratings. The mean avg rating is **3.65**
Technical Implication: The dataset is "positively skewed." This means the model may be biased toward predicting higher ratings. We must ensure the loss function (MSE) penalizes large errors on the rare 1-star ratings to maintain balance.
Business Implication: Users are generally satisfied with the content they choose to rate. This suggests a "selection bias"—users usually rate movies they liked.



Observation: The distribution of movie ratings exhibits a heavy-tail characteristic. While the mean number of ratings per movie is approximately 5,654, the median is significantly lower, indicating that a small number of popular titles dominate the dataset. We implemented a threshold of 100 ratings to exclude the 'long tail' of rare movies, which reduces noise and prevents overfitting during the embedding learning process.

Key Insights

1 Candidate pool is the ranking ceiling

Expanding N_CANDIDATES from 1,000 to 2,000 cost ~70s extra but raised MAP from 0.082 to 0.087 for two tower + lightbm and 0.04 to 0.05 for WALS + BPR,
The Impact: Giving the reranker a wider retrieval pool allowed it to surface high-quality "hidden gems" that were previously choked at the stage 1 boundary.
Engineering Win: Scaling to 2,000 candidates generated training matrices exceeding 30 million rows. We defeated Out-of-Memory (OOM) crashes by bypassing Pandas dataframes, downcasting features to float16, and aggressively triggering manual garbage collection.

2 Tree-Based Reranking Outperforms Linear BPR

Insight: Changing from a linear BPR ranker to a gradient-boosted decision tree (LGBMRanker) caused our largest single leap in ranking quality.
The Culprit: The initial baseline transition from WALS to a PyTorch Two-Tower model saw a performance dip (~0.04–0.05 MAP@10) because the linear BPR ranker could not effectively map the complex, non-linear embedding structures outputted by deep learning towers.
The Fix: LightGBM naturally handles multi-modal feature interactions, blending raw 32-dimensional user/item embeddings with behavioral features.

3 Feature Engineering Captures Short-Term Intent

Insight: Raw embeddings capture static user preferences, but explicit feature engineering captures real-time context. The **dense features**, show two critical behavioral signals:

`recency_score`: Measures the cosine similarity between a candidate movie and the *very last movie* the user watched. This captured immediate, short-term session inten which dynamically shifts recommendations over time.

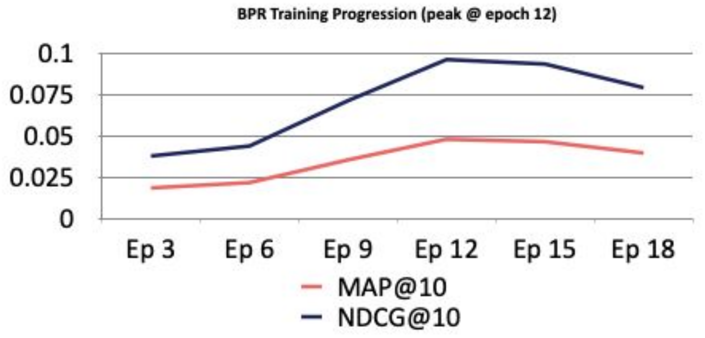
`rating_gap`: Calculated by subtracting a user's historical average rating from a candidate's average rating. This taught LightGBM to actively penalize items likely to fall below that individual user's personal quality threshold.

4 Explicit Feedback Overcomes Bias Mismatch

Insight: Optimizing the Stage 1 Two-Tower retrieval model with a Mean Squared Error (MSE) objective over explicit star ratings (1–5) provided a far more stable foundational retrieval signal than typical implicit feedback alternatives.
Structural Safety: By explicitly incorporating user_bias, item_bias, and global_bias scalar parameters directly into the PyTorch network architecture, the system safely falls back to robust statistical averages when encountering sparse data or zero-history profiles.

Experimental Results

Model	MAP@10	RMSE
Baseline : WALS + BPR (N=1,000)	0.04	~1.003
WALS + BPR (N=2,000)	0.0.05	~1.003
Two tower + LightGBM (N=1,000)	0.082	0.9833
Two tower + LightGBM (N=2,000) ✓	0.087	0.9833



Expanding N is a cheap way to raise recall before improving the ranker itself.

Model Architecture Leap (+118% Gain)

Transitioning the entire core system from the legacy WALS + BPR pipeline (MAP@10 = 0.040) to the modern Two-Tower + LightGBM pipeline (MAP@10 = 0.087) yielded a **118% net relative improvement** in recommendation accuracy.

Cheap Gains via Candidate Pool Expansion

The Recall Ceiling: In the WALS model, expanding the candidate pool size from N=1,000 to N=2,000 boosted the recall ceiling of test positives from **~47% to ~61%**.

The Downstream Payoff: When applied to our final LightGBM architecture, this expansion gave the tree ranker a richer field of hidden gems to evaluate, scaling the final metric directly from **0.082 to 0.087**.

Core Finding:
Rating Prediction ≠ Ranking Quality

Optimizing early-stage models purely for low rating error (RMSE) does not automatically yield high-quality personalized feeds. True ranking performance requires a dedicated, pairwise learning-to-rank optimization layer (LambdaMART objective) fed with context-aware, engineered features like recency_score and rating_gap.

Comparison of Sample Recommendations (Warm User) (cust_id = 79)

Pipeline: WALS top-2000 → filter seen → BPR rerank → top-10

Rank	Title	Yr	WALS★	BPR
1	Ask This Old House: Season 1 Recommended because it's from the 2000s, your most-enjoyed era; a	2002	4.45	4.3776
2	MI-5: Vol. 2 Recommended because it's from the 2000s, your most-enjoyed era; a	2003	4.44	4.3311
3	One Good Cop Recommended because it's from the 1990s, a decade you've enjoyed;	1991	4.44	4.2813
4	House II: The Second Story Recommended because it's from the 1980s, a decade you've enjoyed;	1987	4.78	4.2092
5	The Thing Recommended because it's from the 1980s, a decade you've enjoyed;	1982	4.56	4.0085
6	Veer-Zaara Recommended because it's from the 2000s, your most-enjoyed era; a	2004	4.60	4.0056
7	Critters Recommended because it's from the 1980s, a decade you've enjoyed;	1986	4.56	3.9575
8	Suicide Club Recommended because it's from the 2000s, your most-enjoyed era; a	2002	4.45	3.9405
9	Doobie Brothers: Rockin' Down the Highwa Recommended because it's from the 2000s, your most-enjoyed era; a	2004	4.46	3.9045
10	Red Shoe Diaries: Hotline Recommended because it's from the 1990s, a decade you've enjoyed;	1994	4.60	3.9041

Pipeline: Two Tower top-2000 → filter seen → LIGHTGBM rerank → top-10

#1	Shark Tale (2004)	2.3758	Customer 784621 (5★, Sim: 0.95)	Customer 972103 (4★, Sim: 0.94)
#2	Die Another Day (2002)	2.3187	Customer 784621 (4★, Sim: 0.95)	Customer 2588720 (4★, Sim: 0.93)
#3	Austin Powers (1999)	2.1930	Customer 2577378 (4★, Sim: 0.94)	Customer 981753 (4★, Sim: 0.92)
#4	The Terminal (2004)	2.0539	Customer 1458835 (4★, Sim: 0.93)	Customer 981753 (5★, Sim: 0.92)
#5	Elf (2003)	2.0405	Customer 611251 (4★, Sim: 0.94)	Customer 260260 (5★, Sim: 0.93)
#6	Dr. Dolittle (1998)	1.9831	Customer 260260 (4★, Sim: 0.93)	Customer 981753 (4★, Sim: 0.92)
#7	Ransom (1996)	1.8562	Customer 784621 (4★, Sim: 0.95)	Customer 1002832 (5★, Sim: 0.95)
#8	Once Upon a Time in Mexico (2003)	1.8257	Customer 784621 (4★, Sim: 0.95)	Customer 981753 (4★, Sim: 0.92)
#9	Mrs. Doubtfire (1993)	1.8221	Customer 2577378 (5★, Sim: 0.94)	Customer 972103 (4★, Sim: 0.94)
#10	Shallow Hal (2001)	1.8087	Customer 290916 (4★, Sim: 0.91)	Customer 1522324 (5★, Sim: 0.90)

Cold users (40% of test set) get a fallback list ranked purely by biases : identical, globally popular high-rated items for every cold user.

Future Improvements

Content-based cold-start features

Action: Integrate genre, cast, and release year metadata directly into the LightGBM ranker.
Value: Targets the critical 89.1% cold-user cohort where collaborative embeddings lack history.

High impact

Larger candidate pool (N=5,000)

Action: Expand the Two-Tower retrieval cutoff from 2,000 to 5,000 items.
Value: Raises the recall ceiling to ~78% with minimal compute overhead.

Low complexity

User neighbourhood (kNN on two tower factors)

Action: Compute cluster neighborhoods using the 32-dimensional user embeddings.
Value: Provides a strong fallback collaborative signal for cold or low-activity profiles.

High impact

Multi-objective loss (Joint MSE + LambdaMART)

Action: Train a single end-to-end network optimizing both rating precision and pairwise ranking simultaneously.
Value: Eliminates alignment mismatches between the retrieval and reranking stages.

High complexity

Real time online learning

Action: Implement streaming model updates to refresh user/item vectors upon new ratings without full nightly retrains.
Value: Captures immediate session-to-session drift in user preferences on the fly.

Very high complexity

THANK YOU!

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